CHEONGJAE LEE
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EDUCATION

BOSTON UNIVERSITY — Boston, MA
Master's in Electrical Engineering — Expected Graduation: 12/27
Main Focus of study: ***Cyber-Physical Systems, Robotics, Computer Vision***

THE PENNSYLVANIA STATE UNIVERSITY — State College, PA
Bachelor of Science in ***Electrical Engineering*** — 12/23
Minor in ***Engineering Mechanics***

PROJECTS

MANIMO: DISASTER-RESPONSE MOBILE MANIPULATOR — Boston University
EC601 Team Project (in progress) — Fall 2026 – Present

- Developed a ROS 2 (Python) autonomy node for a simulated wheeled mobile manipulator with a Franka Panda arm in NVIDIA Isaac Sim, completing an autonomous pick-and-place mission between tables.
- Fused 2D LiDAR clustering with camera color detection in a 360° survey to find the table holding the target.
- Implemented precision docking from a LiDAR line fit of the table face, with a LiDAR emergency stop.

MULTI-CROP DETECTION FOR AN AGRICULTURAL ROBOT — Boston University
Robotics Club, Agrobot Team (in progress) — Fall 2026 – Present

- Built a PyTorch pipeline that converts the LaboroTomato dataset (804 images, 10,610 polygon labels) to bounding boxes with zero label loss and fine-tunes a COCO-pretrained Faster R-CNN (ResNet-50 FPN).
- Reached 0.90 precision and 0.91 recall (F1 0.91, IoU > 0.5) on the validation set.
- Diagnosed small-object misses (17% of tomatoes, 58% of misses) with size-bucketed recall; a size-aware threshold raised small-tomato recall from 0.72 to 0.81.

PROFESSIONAL EXPERIENCE

SOELECT INCORPORATED — Greensboro, NC
Manufacturing Process Engineer — 03/26 – 05/26

- Ran roll-to-roll lithium metal anode production and evaluated anode quality using optical microscopy, contact-angle measurement, peel-strength testing, and SEM/EDX.
- Assembled ~5 mAh pouch cells and assessed performance through EIS, cycling tests, and teardown analysis.

FASTLION ENERGY INCORPORATED — State College, PA
Electrical Engineer — 07/24 – 03/26

- Designed and prototyped lithium-ion Battery Management Systems for 6T military and industrial battery platforms, covering cell monitoring, fault detection, and system-level protection.
- Designed multi-layer PCBs with PWM control, precision analog sensing, SPI/I2C, and high-current power paths.
- Developed and validated CAN communication protocols for battery telemetry, diagnostics, and control integration with vehicle and system controllers.
- Brought up, debugged, and validated prototype hardware with oscilloscopes, logic analyzers, and bench tools.

EC POWER GROUP — State College, PA
Electrical Engineer (promoted from Electrical Engineering Intern, 02/24) — 05/23 – 07/24

- Led end-to-end development of a custom cell-level battery test platform, spanning PCB design, embedded C/C++ firmware, sensor interfaces, and data acquisition.
- Designed and programmed microcontroller-based systems for Battery Management System prototyping.
- Developed firmware and PCB layouts for an automated battery test rig, and debugged prototype circuits.

SKILLS

Programming: Python, C/C++, MATLAB, LabVIEW
Robotics and ML: ROS 2, NVIDIA Isaac Sim, PyTorch, torchvision, Linux (WSL)
Hardware: Altium, KiCad, multi-layer PCB design, CAN, SPI, I2C, Arduino, oscilloscopes, logic analyzers
Lab and Mechanical: SEM/EDX, EIS, li-ion battery cycle testing, SolidWorks, OnShape, 3D printing